MEMORANDUM

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TO: Joseph Pinheiro, Professor, Mechanical Engineering

FROM:

Eduardo Garcia	Apolonio Ordaz
Egarc217@calpoly.edu	Apordaz@calpoly.edu
Gabrieljr Lemus	Caleb Bradbury
Glemus02@calpoly.edu	Cabradbu@calpoly.edu

Team: Spades
Copies: 1
SUBJECT: **Lab 06 – Boundary Layer Flow Over a Plate**

Abstract

The purpose of this experiment was to investigate the development of the boundary layer over a flat plate in a low-speed wind tunnel. Streamwise velocity profiles were measured at downstream distances of 11.5 cm and 21.5cm from the leading edge using a Pitot static tube and manometer. Experimental pressure readings were converted to local velocities using Bernoulli's equation, and the resulting velocities were used to determine the boundary layer thickness, displacement thickness, momentum thickness and drag coefficient using numerical integration, via the trapezoid rule. The experimental boundary layer thickness increased from 2.775 mm to 4.48 mm as the distance from the leading edge increased, confirming boundary layer grows as it moves downstream. When comparing with Blasius exact solutions for laminar boundary layers and turbulent power law predictions, the measured boundary layer thicknesses fell between laminar and turbulent values. However, the momentum thickness, drag coefficient, and drag forces were closer to the laminar predictions. Overall, the results suggest that the boundary layer was transitional rather than fully laminar or fully turbulent.

Background/Theory

When a viscous fluid flows over a solid surface, the no-slip condition at the wall forces the fluid velocity to zero at the surface. Moving away from the wall, the velocity gradually recovers until it matches the free-stream value. This thin region of velocity transition is known as the boundary layer, and its thickness grows with increasing distance downstream from the leading edge of the plate. The nature of the boundary layer whether laminar or turbulent is governed by the local Reynolds number, which represents the ratio of inertial to viscous forces at a given downstream location. For flow over a flat plate, the boundary layer is expected to remain laminar when the local Reynolds number is below approximately 5×10^5, beyond which transition to turbulent flow occurs. See (figure 1)

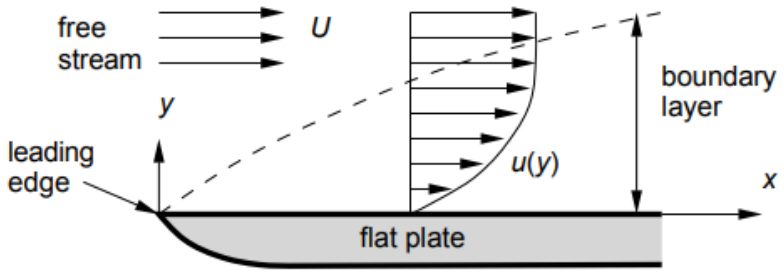


Figure 1. Boundary layer for uniform flow over a flat plate.

For a laminar boundary layer, the Blasius solution to the boundary layer equations predicts that the thickness grows proportionally to the square root of the downstream distance and inversely with the square root of the local Reynolds number. The boundary layer thickness is formally defined as the wall-normal distance at which the local velocity reaches 99% of the free-stream value. For turbulent boundary layers, an empirical power-law correlation predicts a faster rate of growth, scaling with the Reynolds number raised to the negative one-fifth power.

Two additional integral quantities provide a more complete description of the boundary layer. The displacement thickness δ^* represents the distance by which the outer streamlines are effectively pushed away from the wall due to the slow-moving fluid near the surface in other words, it is the thickness of an imaginary zero-velocity layer that would produce the same reduction in mass flow as the actual boundary layer. The momentum Θ thickness similarly quantifies the loss of momentum flux caused by the boundary layer, and it is computed by integrating the product of the local velocity ratio and the velocity deficit across the boundary layer. Both quantities are smaller than the boundary layer thickness itself, but they carry more physical significance for aerodynamic analysis. Theoretical correlations for laminar flow predict that the displacement thickness scales with the inverse square root of the Reynolds number, while turbulent predictions follow the one-fifth power law.

The momentum thickness is directly connected to the aerodynamic drag on the plate through the von Kármán momentum integral relation, which states that the overall drag coefficient is equal to twice the momentum thickness divided by the distance from the leading edge. The total viscous drag force is then obtained by multiplying the momentum thickness by the air density, the plate width, and the square of the free-stream velocity. These relationships allow the drag to be estimated entirely from the measured velocity profile without directly measuring the wall shear stress.

Apparatus and Experimental Procedure

The experiment was conducted using a low-speed open-circuit wind tunnel with a flat plate mounted parallel to the oncoming flow. The plate was oriented so that its leading edge defined the origin of boundary layer growth. Before testing began, the ambient temperature and atmospheric pressure were recorded to calculate air density from the ideal gas law, which was needed for subsequent velocity and force calculations.

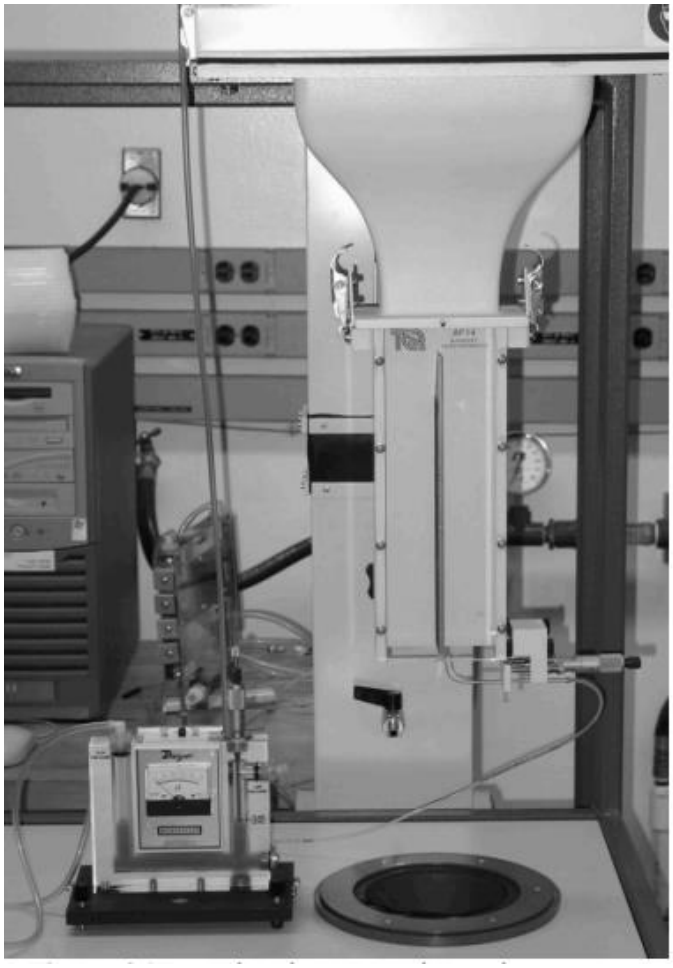


Figure 2. Experimental Apparatus

Velocity profiles within the boundary layer were measured using a Pitot-static tube connected to an inclined water manometer. The Pitot tube was mounted on a traversing mechanism that allowed it to be positioned at precise distances from the plate surface, perpendicular to the flow. Measurements were collected at two streamwise stations located 11.5 cm and 21.5 cm from the leading edge of the plate. At each station, the Pitot tube was stepped outward from the plate surface in small increments using a micrometer, and the manometer deflection, measured in inches of water, was recorded at each position.

The manometer readings were converted to dynamic pressure in Pascals, and local stream velocities were then calculated from the Bernoulli equation applied between the stagnation point at the Pitot tube tip and the static reference. The traversal at each station continued until the recorded velocity stopped increasing and remained constant over several consecutive positions, indicating that the probe had exited the boundary layer and entered the free stream. The free-stream velocity was taken as this plateau value, and the boundary layer thickness was identified as the wall-normal position at which the local velocity reached 99% of the free stream velocity.

Once the velocity profiles were established, the displacement thickness and momentum thickness at each station were computed numerically using the trapezoidal rule. A data point corresponding to zero velocity

at the wall was included in each profile to enforce the no-slip condition and capture the contribution of the near-wall region. The drag coefficient was then calculated from the momentum thickness using the von Kármán relation, and the drag force was computed from the air density, plate width, free-stream velocity, and momentum thickness. All experimental values were compared against the theoretical predictions for both laminar (Blasius) and turbulent (power-law) boundary layers.

Analysis

Velocity Profile

The pressure measured by the Pitot tube was converted into stream velocity using the Bernoulli based equation.

$$u = \sqrt{\frac{2\Delta P}{\rho}} \tag{1}$$

where ΔP is the measured dynamic pressure from the manometer converted from in H20 to Pascals and ρ is the density of the air at the measured atmosphere pressure and temperature. The calculated velocities were then normalized to the free stream velocity, *U,* which was calculated by also using equation 1. The normalized distance from the plate was calculated as $\frac{y}{\delta}$. δ was the experimentally determined boundary layer which was taken to be the thickness corresponding to $\frac{u}{U} \approx 0.99$.

The measured value of δ was compared with the theoretical laminar and turbulent correlations provided in the lab manual. For laminar flow, the Blasius solution predicts:

$$\frac{\delta}{x} = \frac{5.0}{\sqrt{Re_x}} \tag{2}$$

Where the assumptions of 2-D flow, steady state, incompressible flow, no pressure gradients, and an assumed polynomial velocity profile were assumed.

For turbulent flow, the following empirical formula was used:

$$\frac{\delta}{x} = \frac{0.370}{Re_x^{1/5}} \tag{3}$$

Where the assumptions of steady flow, 2-D flow, incompressible, no body forces parallel to the flow, and a velocity profile of power law one-seventh were assumed.

Displacement thickness

The displacement thickness was determined numerically using the trapezoid rule applied to the measured velocity profile. The displacement thickness equation is as follows.

$$\delta^* = \int_0^\infty \left(1 - \frac{u}{U}\right) dy \tag{4}$$

The term (1-u/U) was calculated for each measured location and multiplied by the corresponding space between data points to determine each partial integral. The sum of each partial integral yielded the experimental displacement thickness.

The experimental displacement thickness was compared with the theoretical predictions. For laminar flow:

$$\frac{\delta^*}{x} = \frac{1.721}{\sqrt{Re_x}} \tag{5}$$

For turbulent flow:

$$\frac{\delta^*}{x} = \frac{0.0463}{Re_x^{1/5}} \tag{6}$$

Momentum Thickness

The momentum thickness was calculated in the same manner as the displacement thickness using the measured velocity profile and the trapezoidal rule.

$$\Theta = \int_0^\infty \frac{u}{U}\left(1 - \frac{u}{U}\right) dy \tag{7}$$

The quantity $\frac{u}{U}\left(1 - \frac{u}{U}\right)$ was evaluated multiplied by the space between data points to determine each partial integral. These partial integrals were the summed to obtain the experimental momentum thickness. The experimentally determined momentum thickness was compared with theoretical values predicted for laminar and turbulent boundary layers. For laminar flow:

$$\frac{\Theta}{\mathrm{x}} = \frac{0.664}{\sqrt{\mathrm{Re_x}}} \tag{8}$$

For turbulent flow:

$$\frac{\Theta}{x} = \frac{0.0360}{Re_x^{1/5}} \tag{9}$$

Drag Coefficient and Drag Force

The Drag force coefficient was determined experimentally by using the calculated momentum thickness and using the von Kármán momentum integral.

$$C_D = \frac{2\Theta}{L} \tag{10}$$

Where L is the distance from the leading edge to the measured location.

The drag force on the plate due to the shear stress was then calculated using the following:

$$F_D = \rho b U^2 \Theta \tag{11}$$

Where b is the plate width and ρ was the density of the air. The experimental drag force and drag coefficient were compared to the theoretical values expected for both laminar and turbulent flow over a flat plate.

Results

Table 1. Experimental Boundary Layer Analysis (x = 11.5 cm)

Measurement	Distance From Plate, y (mm)	Pitot Tube Pressure $(mm\ H_2O)$	Pressure, P (Pa)	Stream-Wise Velocity, u (m/s)
1	0.25	3.81	37.3	7.92
2	0.31	4.953	48.5	9.03
3	0.37	7.112	69.7	10.82
4	0.43	9.271	90.8	12.35
5	0.48	10.922	107	13.41
6	0.54	13.081	128.1	14.67
7	0.6	14.859	145.5	15.64
8	0.66	16.002	156.7	16.23
9	0.71	17.018	166.7	16.74
10	0.77	18.288	179.1	17.35
11	0.82	19.05	186.6	17.71
12	0.88	20.32	199	18.29
13	0.93	20.828	204	18.52
14	1.17	21.7932	213.5	18.94
15	1.4	22.987	225.2	19.45
16	1.48	23.241	227.7	19.56
17	1.57	23.368	228.9	19.61
18	1.65	23.495	230.1	19.66
19	2.78	23.876	233.9	19.82
20	3.90	23.876	233.9	19.82

Table 1 shows the collected and calculated values measured at 11.5cm from the leading edge of the flat plate. The table includes the distance from the plate, the Pitot tube pressure converted to mm H_2O, the calculated pressure difference, and the resulting stream wise velocity. The experimental boundary layer

was estimated to be about 2.78 mm. The local velocity near the edge of the boundary layer was calculated to be about 99% of our measured free stream velocity of about 20.03 m/s, resulting in a value for *u* equal to 19.82 m/s. Using this *u* value, we found the related distance from the plate to be our boundary layer of about 2.78 mm. The final distance of 3.90 mm had the same velocity and was used as confirmation that we had reached the free stream rather than the actual boundary layer thickness. The same procedure was repeated for the 21.5 cm location, and the corresponding table is included in the attachment section as "Experimental Boundary Layer Analysis (x = 21.5 cm)."

Table 2. Normalized Velocity Profile and Thickness Integrals (x = 11.5 cm)

Normalized Distance, (y/δ)	**Normalized Velocity, (u/U)**	**(1 – *u/U*)**	**(*u/U*) (1-*u/U*)**	**Displacement Thickness Partial, δ^* (mm)**	**Momentum Thickness Partial, Θ (mm)**
0.000	0.000	1.000	0.000	---	---
0.090	0.395	0.605	0.239	0.201	0.030
0.112	0.451	0.549	0.247	0.035	0.015
0.133	0.540	0.460	0.248	0.030	0.015
0.155	0.616	0.384	0.237	0.025	0.015
0.173	0.669	0.331	0.221	0.018	0.011
0.195	0.732	0.268	0.196	0.018	0.013
0.216	0.781	0.219	0.171	0.015	0.011
0.238	0.810	0.190	0.154	0.012	0.010
0.256	0.836	0.164	0.137	0.009	0.007
0.277	0.866	0.134	0.116	0.009	0.008
0.295	0.884	0.116	0.103	0.006	0.005
0.317	0.913	0.087	0.079	0.006	0.005
0.335	0.924	0.076	0.070	0.004	0.004
0.422	0.945	0.055	0.052	0.016	0.015
0.505	0.971	0.029	0.028	0.010	0.009
0.533	0.976	0.024	0.023	0.002	0.002
0.566	0.981	0.019	0.019	0.002	0.002
0.595	0.981	0.019	0.018	0.002	0.002
1.000	0.989	0.011	0.011	0.017	0.016
SUM	**---**	**---**	**---**	**0.435**	**0.194**

Table 2 shows the normalized velocity profile and the partial integral values that we used to calculate both displacement and momentum thickness at x = 11.5 cm. We included the wall point (0, 0) to account for the area between the plate and the first Pitot tube reading. The trapezoidal rule was used to calculate the partial integrals for displacement and momentum thickness. Summing the results gave $\delta^* =$ .435mm and $\Theta =$ .194mm respectively. We repeated the same process for the x = 21.5 cm location, resulting in the values $\delta^* =$ 0.582mm and $\Theta =$ 0.330mm. The corresponding table is included in the attachment section as "Normalized Velocity Profile and Thickness Integral (x = 21.5 cm)."

The sample calculations for x = 11.5 cm show how the normalized and partial integrals were obtained.

$$\frac{y}{\delta} = \frac{.25mm}{2.775mm} = .09$$

$$\frac{u}{U} = \frac{7.92m/s}{20.03m/s} = .395$$

$$1 - \frac{u}{U} = 1 - .395 = .605$$

$$\left(\frac{u}{U}\right)\left(1 - \frac{u}{U}\right) = .395 * .605 = .239$$

$$\delta^* = \left(\frac{.605 + 1.000}{2}\right)(.25 - 0) = .201mm$$

$$\Theta = \left(\frac{.239 + 0}{2}\right)(.25 - 0) = .03mm$$

Table 3. Experimental and Theoretical Boundary Layer Results (x = 11.5 cm)

Parameter	Experimental	Theoretical Laminar (Blasius)	Theoretical Turbulent (Power Law)
Boundary Layer Thickness, $\delta\ (mm)$	2.775	1.487	3.926
Displacement Thickness, $\delta^*\ (mm)$	0.435	0.512	0.491
Momentum Thickness, $\Theta\ (mm)$	0.194	0.197	0.382
Drag Coefficient, C_D	0.00337	0.00343	0.00664
Drag Force, $F_D\ (N)$	0.00463	0.00472	0.00912

Table 4. Experimental and Theoretical Boundary Layer Results (x = 21.5cm)

Parameter	Experimental	Theoretical Laminar (Blasius)	Theoretical Turbulent (Power Law)
Boundary Layer Thickness, $\delta\ (mm)$	4.48	2.033	6.476
Displacement Thickness, $\delta^*\ (mm)$	0.582	0.7	0.81
Momentum Thickness, $\Theta\ (mm)$	0.33	0.27	0.63
Drag Coefficient, C_D	0.00307	0.00251	0.00586
Drag Force, $F_D\ (N)$	0.00788	0.00645	0.01505

Tables 3 and 4 compare the experimental boundary layer results with the theoretical laminar Blasius and theoretical turbulent correlations. The experimental boundary layer thickness, displacement thickness, and momentum thickness were all obtained from the measured velocities and the trapezoidal rule's integration. The coefficient for drag and the drag force were calculated using Equations (10) and (11), respectively. From the data we see that the experimental boundary layers were thicker than the laminar predictions but smaller than the turbulent predictions, indicating that our boundary layer was not entirely laminar. The experimental momentum thickness, drag coefficient, and drag force values were closer to the laminar predictions, suggesting that the flow may not have been fully turbulent and the near wall Pitot tube measurement uncertainty affected the results. The drag forces we calculated were small; this makes sense because the plate width was small and the boundary layer momentum thickness was calculated to be lass the 1mm.

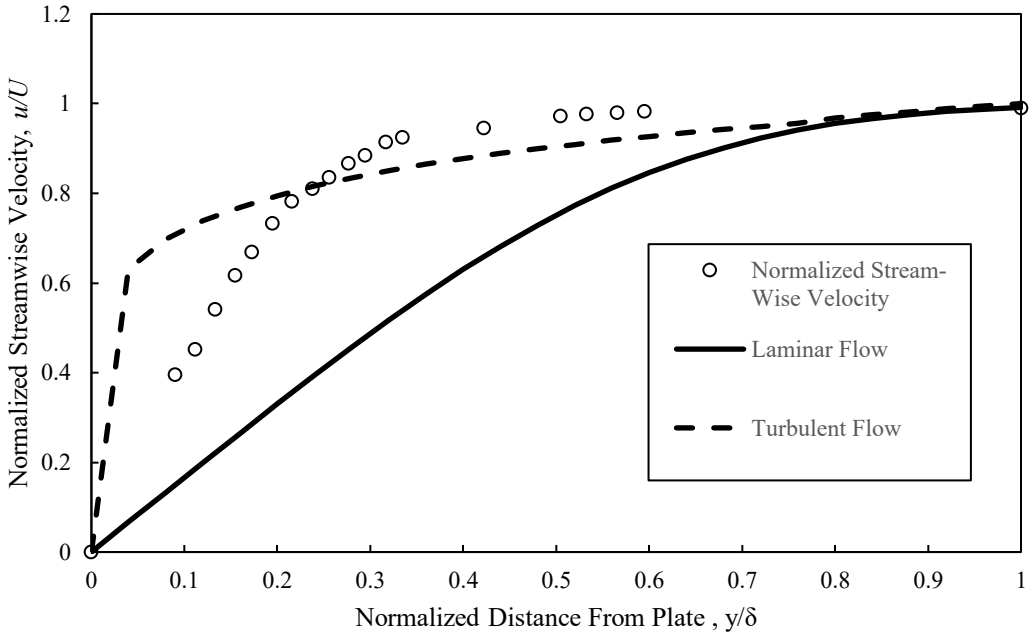


Figure 3. Experimental and Theoretical Normalized Velocity Profile (x = 11.5 cm)

In Figure 3 we see the comparison between the experimental normalized velocity profile, the laminar Blasius profile, and the turbulent power law profile, all at x = 11.5 cm. The experimental data rises faster than the laminar prediction, being a closer representation of the turbulent curve over much of the profile, this suggests that the boundary layer was not solely laminar. It is important to note that some of our experimental points were above the turbulent prediction, this may have been caused by the uncertainty in the near wall Pitot tube measurement and the wind tunnel turning off during the experiment. While not an ideal profile follows the laminar predictions, we can still say that the profile suggests there was a transitional section rather than a fully laminar or fully turbulent boundary layer.

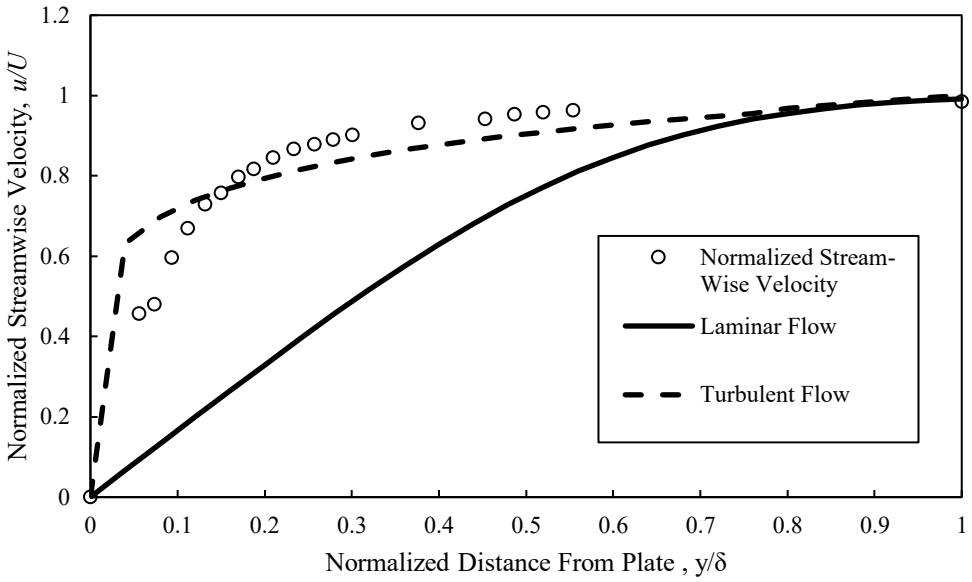


Figure 4. Experimental and Theoretical Normalized Velocity Profile (x = 21.5 cm)

Figure 4 compares the experimental normalized velocity profile, the laminar Blasius profile, and the turbulent power law profile at x = 21.5 cm. The experimental data follows the turbulent profile closer than the laminar profile, especially through the middle of the boundary layer. This is a reasonable pattern because the flow is farther downstream along the plate, which gives the boundary layer more distance to develop and transition. Much like Figure 3, our profile does not perfectly match either curve, this is likely because of the Pitot tube measurement uncertainty. Overall, the profile suggests transitional behavior and that it follows a similar profile to that of theoretical turbulent flow.

Conclusion

The results showed that the boundary layer increased in thickness as the flow moved downstream, growing from 2.775 mm at $x = 11.5$cm to 4.48 mm at $x = 21.5$cm. This trend agrees with the expected theory since viscous effects spread farther from the wall as distance from the leading-edge increases.

When compared with the theoretical laminar Blasius solution and turbulent power-law correlation, the experimental boundary layer thicknesses fell between the laminar and turbulent predictions. At $x = 11.5$cm, the experimental boundary layer thickness was 2.775 mm, compared to 1.487 mm for laminar flow and 3.926 mm for turbulent flow. At $x = 21.5$cm, the experimental value was 4.48 mm, compared to 2.033 mm for laminar flow and 6.476 mm for turbulent flow. These results suggest that the flow was not fully laminar or fully turbulent but instead was likely in a transitional state. The Reynolds numbers at 11.5 cm and 21.5 cm from the leading edge were approximately $1.50 \times 10^5$and 2.80×10^5, respectively. Although both values are below the typical transition Reynolds number of $5 \times 10^5$for flow over a flat plate, factors such as surface roughness, small disturbances in the wind tunnel flow, and measurement uncertainty can cause transition to begin earlier than expected. The normalized velocity profile plots also support this conclusion, since the experimental curves generally followed the turbulent profile more closely than the laminar profile, especially farther downstream, while still not matching the turbulent

model exactly. Overall, the experimental results indicate that the boundary layer was in the transitional region rather than fully laminar or fully turbulent.
The displacement thickness, momentum thickness, drag coefficient, and drag force also increased downstream. The displacement thickness increased from 0.435 mm to 0.582 mm, while the momentum thickness increased from 0.194 mm to 0.330 mm. The drag force increased from 0.00463 N at $x = 11.5\text{cm}$ to 0.00788 N at $x = 21.5\text{cm}$. Although the experimental boundary layer thicknesses were between laminar and turbulent predictions, the momentum thickness, drag coefficient, and drag force were closer to the laminar values, indicating that the flow had not become fully turbulent.

Possible sources of error include uncertainty in positioning the Pitot tube close to the plate, difficulty reading small manometer pressure changes, slight nonuniformity in the free stream, and disturbances caused by the wind tunnel turning off during the experiment. Near-wall measurement error was especially important because the velocity gradient is largest close to the plate, so small errors in this region can affect the calculated displacement and momentum thicknesses. Overall, the experiment successfully demonstrated boundary layer growth over a flat plate and showed that the measured flow was most likely transitional, with boundary layer thicknesses between laminar and turbulent theory but drag-related quantities closer to laminar predictions.

References

[1] ME 347 Lab Manual

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Attachments

1. Experimental Boundary Layer Analysis (x = 21.5 cm)
2. Normalized Velocity Profile and Thickness Integrals (x = 21.5 cm)
3. EES Prelab Calculations

Attachment 1 - Experimental Boundary Layer Analysis (x = 21.5 cm)

Measurement	Distance From Plate, y (mm)	Pitot Tube Pressure $(mm\ H_2O)$	Pressure, P (Pa)	Stream-Wise Velocity, u (m/s)
1	0.25	5.08	49.8	9.14
2	0.33	5.588	54.7	9.59
3	0.42	8.636	84.6	11.92
4	0.5	10.922	107	13.41
5	0.59	12.954	126.9	14.6
6	0.67	13.97	136.8	15.16
7	0.76	15.494	151.8	15.97
8	0.84	16.256	159.2	16.36
9	0.94	17.399	170.4	16.92
10	1.05	18.288	179.1	17.35
11	1.15	18.796	184.1	17.59
12	1.25	19.304	189.1	17.83
13	1.35	19.812	194.1	18.06
14	1.69	21.082	206.5	18.63
15	2.03	21.59	211.5	18.85
16	2.18	22.098	216.5	19.07
17	2.33	22.352	218.9	19.18
18	2.48	22.606	221.4	19.29
19	4.48	23.622	231.4	19.72
20	6.48	23.876	233.9	19.82

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Attachment 2 - Normalized Velocity Profile and Thickness Integrals (x = 21.5 cm)

Normalized Distance, (y/δ)	Normalized Velocity, (u/U)	$(1 - u/U)$	$(u/U)(1-u/U)$	Displacement Thickness Partial, δ^* (mm)	Momentum Thickness Partial, Θ (mm)
0	0	1	0	—	—
0.056	0.456	0.544	0.248	0.193	0.031
0.074	0.479	0.521	0.25	0.043	0.02
0.094	0.595	0.405	0.241	0.042	0.022
0.112	0.669	0.331	0.221	0.029	0.018
0.132	0.729	0.271	0.198	0.027	0.019
0.15	0.757	0.243	0.184	0.021	0.015
0.17	0.797	0.203	0.162	0.02	0.016
0.188	0.817	0.183	0.149	0.015	0.012
0.21	0.845	0.155	0.131	0.017	0.014
0.234	0.866	0.134	0.116	0.016	0.014
0.257	0.878	0.122	0.107	0.013	0.011
0.279	0.89	0.11	0.098	0.012	0.01
0.301	0.901	0.099	0.089	0.01	0.009
0.377	0.93	0.07	0.065	0.029	0.026
0.453	0.941	0.059	0.055	0.022	0.021
0.487	0.952	0.048	0.046	0.008	0.008
0.52	0.957	0.043	0.041	0.007	0.006
0.554	0.963	0.037	0.036	0.006	0.006
1	0.984	0.016	0.015	0.053	0.051
SUM				**0.582**	**0.33**

Attachment 3 – EES Prelab Calculations

